

Template

Studentnames and studentnumbers here

2026-06-15

Set-up your environment

Title Page

Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
#dataset <- midwest  
  
library(readr)  
total_govt_exp <- read_csv("data/total_govt_exp.csv")
```

```
## Rows: 340 Columns: 23
## -- Column specification -----
## Delimiter: ","
## chr (17): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, uni...
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(total_govt_exp)
```

```
library(readr)
general_income_ineq_gini_ <- read_csv("data/general_income_ineq(gini).csv")
```

```
## Rows: 368 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr (13): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, age...
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(general_income_ineq_gini_)
library(readr)
midwest <- read_csv("data/midwest.csv")
```

```
## Rows: 437 Columns: 28
## -- Column specification -----
## Delimiter: ","
## chr (3): county, state, category
## dbl (25): PID, area, poptotal, popdensity, popwhite, popblack, popamerindian...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(midwest)
```

```
general_income_ineq_gini_$geo
```

```
## [1] "AL"      "AL"      "AL"      "AL"      "AL"      "AL"
## [7] "AL"      "AT"      "AT"      "AT"      "AT"      "AT"
## [13] "AT"      "AT"      "AT"      "AT"      "AT"      "BE"
## [19] "BE"      "BE"      "BE"      "BE"      "BE"      "BE"
## [25] "BE"      "BE"      "BE"      "BG"      "BG"      "BG"
## [31] "BG"      "BG"      "BG"      "BG"      "BG"      "BG"
## [37] "BG"      "CH"      "CH"      "CH"      "CH"      "CH"
## [43] "CH"      "CH"      "CH"      "CH"      "CH"      "CY"
## [49] "CY"      "CY"      "CY"      "CY"      "CY"      "CY"
```

##	[55]	"CY"	"CY"	"CY"	"CZ"	"CZ"	"CZ"
##	[61]	"CZ"	"CZ"	"CZ"	"CZ"	"CZ"	"CZ"
##	[67]	"CZ"	"DE"	"DE"	"DE"	"DE"	"DE"
##	[73]	"DE"	"DE"	"DE"	"DE"	"DE"	"DK"
##	[79]	"DK"	"DK"	"DK"	"DK"	"DK"	"DK"
##	[85]	"DK"	"DK"	"DK"	"EA20"	"EA20"	"EA20"
##	[91]	"EA20"	"EA20"	"EA20"	"EA20"	"EA20"	"EA20"
##	[97]	"EA20"	"EE"	"EE"	"EE"	"EE"	"EE"
##	[103]	"EE"	"EE"	"EE"	"EE"	"EE"	"EL"
##	[109]	"EL"	"EL"	"EL"	"EL"	"EL"	"EL"
##	[115]	"EL"	"EL"	"EL"	"ES"	"ES"	"ES"
##	[121]	"ES"	"ES"	"ES"	"ES"	"ES"	"ES"
##	[127]	"ES"	"EU27_2020"	"EU27_2020"	"EU27_2020"	"EU27_2020"	"EU27_2020"
##	[133]	"EU27_2020"	"EU27_2020"	"EU27_2020"	"EU27_2020"	"EU27_2020"	"FI"
##	[139]	"FI"	"FI"	"FI"	"FI"	"FI"	"FI"
##	[145]	"FI"	"FI"	"FI"	"FR"	"FR"	"FR"
##	[151]	"FR"	"FR"	"FR"	"FR"	"FR"	"FR"
##	[157]	"FR"	"HR"	"HR"	"HR"	"HR"	"HR"
##	[163]	"HR"	"HR"	"HR"	"HR"	"HR"	"HU"
##	[169]	"HU"	"HU"	"HU"	"HU"	"HU"	"HU"
##	[175]	"HU"	"HU"	"HU"	"IE"	"IE"	"IE"
##	[181]	"IE"	"IE"	"IE"	"IE"	"IE"	"IE"
##	[187]	"IE"	"IS"	"IS"	"IS"	"IS"	"IS"
##	[193]	"IS"	"IS"	"IT"	"IT"	"IT"	"IT"
##	[199]	"IT"	"IT"	"IT"	"IT"	"IT"	"IT"
##	[205]	"LT"	"LT"	"LT"	"LT"	"LT"	"LT"
##	[211]	"LT"	"LT"	"LT"	"LT"	"LU"	"LU"
##	[217]	"LU"	"LU"	"LU"	"LU"	"LU"	"LU"
##	[223]	"LU"	"LU"	"LV"	"LV"	"LV"	"LV"
##	[229]	"LV"	"LV"	"LV"	"LV"	"LV"	"LV"
##	[235]	"ME"	"ME"	"ME"	"ME"	"ME"	"ME"
##	[241]	"ME"	"ME"	"MK"	"MK"	"MK"	"MK"
##	[247]	"MK"	"MK"	"MK"	"MK"	"MK"	"MK"
##	[253]	"MT"	"MT"	"MT"	"MT"	"MT"	"MT"
##	[259]	"MT"	"MT"	"MT"	"MT"	"NL"	"NL"
##	[265]	"NL"	"NL"	"NL"	"NL"	"NL"	"NL"
##	[271]	"NL"	"NL"	"NO"	"NO"	"NO"	"NO"
##	[277]	"NO"	"NO"	"NO"	"NO"	"NO"	"NO"
##	[283]	"PL"	"PL"	"PL"	"PL"	"PL"	"PL"
##	[289]	"PL"	"PL"	"PL"	"PL"	"PT"	"PT"
##	[295]	"PT"	"PT"	"PT"	"PT"	"PT"	"PT"
##	[301]	"PT"	"PT"	"RO"	"RO"	"RO"	"RO"
##	[307]	"RO"	"RO"	"RO"	"RO"	"RO"	"RO"
##	[313]	"RS"	"RS"	"RS"	"RS"	"RS"	"RS"
##	[319]	"RS"	"RS"	"RS"	"RS"	"SE"	"SE"
##	[325]	"SE"	"SE"	"SE"	"SE"	"SE"	"SE"
##	[331]	"SE"	"SE"	"SI"	"SI"	"SI"	"SI"
##	[337]	"SI"	"SI"	"SI"	"SI"	"SI"	"SI"
##	[343]	"SK"	"SK"	"SK"	"SK"	"SK"	"SK"
##	[349]	"SK"	"SK"	"SK"	"SK"	"TR"	"TR"
##	[355]	"TR"	"TR"	"TR"	"TR"	"TR"	"TR"
##	[361]	"TR"	"TR"	"UK"	"UK"	"UK"	"UK"
##	[367]	"UK"	"XK"				

```
summary(general_income_ineq_gini_)
```

```
##      STRUCTURE      STRUCTURE_ID  STRUCTURE_NAME      freq
## Length   :368   Length   :368   Length   :368   Length   :368
## N.unique : 1    N.unique : 1    N.unique : 1    N.unique : 1
## N.blank  : 0    N.blank  : 0    N.blank  : 0    N.blank  : 0
## Min.nchar: 8    Min.nchar: 19   Min.nchar: 56   Min.nchar: 1
## Max.nchar: 8    Max.nchar: 19   Max.nchar: 56   Max.nchar: 1
##
##      Time frequency      age      Age class      statinfo
## Length   :368   Length   :368   Length   :368   Length   :368
## N.unique : 1    N.unique : 1    N.unique : 1    N.unique : 1
## N.blank  : 0    N.blank  : 0    N.blank  : 0    N.blank  : 0
## Min.nchar: 6    Min.nchar: 5    Min.nchar: 5    Min.nchar: 8
## Max.nchar: 6    Max.nchar: 5    Max.nchar: 5    Max.nchar: 8
##
##      Statistical information      geo      Geopolitical entity (reporting)
## Length   :368      Length   :368   Length   :368
## N.unique : 1      N.unique : 39   N.unique : 39
## N.blank  : 0      N.blank  : 0    N.blank  : 0
## Min.nchar: 38      Min.nchar: 2    Min.nchar: 5
## Max.nchar: 38      Max.nchar: 9    Max.nchar: 41
##
##      TIME_PERIOD      Time      OBS_VALUE      Observation value
## Min.   :2014      Mode:logical   Min.   :20.90   Mode:logical
## 1st Qu.:2016      NAs :368      1st Qu.:27.20   NAs :368
## Median :2018      Median :30.10
## Mean   :2018      Mean   :30.39
## 3rd Qu.:2021      3rd Qu.:33.00
## Max.   :2023      Max.   :46.60
##
##      OBS_FLAG      Observation status (Flag) V2 structure CONF_STATUS
## Length   :368      Length   :368      Mode:logical
## N.unique : 1      N.unique : 1      NAs :368
## N.blank  : 0      N.blank  : 0
## Min.nchar: 1      Min.nchar: 20
## Max.nchar: 1      Max.nchar: 20
## NAs      :345      NAs      :345
## Confidentiality status (flag)
## Mode:logical
## NAs :368
##
##
##
##
```

```
library(readr)
tertiary_edu_govt_exp <- read_csv("data/tertiary_edu_govt_exp.csv")
```

```
## Rows: 320 Columns: 25
## -- Column specification -----
## Delimiter: ","
## chr (19): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, uni...
```

```
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(tertiary_edu_govt_exp)
library(readr)
gender_income_ineq <- read_csv("data/gender_income_ineq.csv")
```

```
## Rows: 333 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr (13): STRUCTURE, STRUCTURE_ID, STRUCTURE_NAME, freq, Time frequency, uni...
## dbl (2): TIME_PERIOD, OBS_VALUE
## lgl (4): Time, Observation value, CONF_STATUS, Confidentiality status (flag)
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(gender_income_ineq)
```

midwest is an example dataset included in the tidyverse package

```
gender <- read_csv("data/gender_income_ineq.csv")
gender = gender[,c ("geo", "Geopolitical.entity..reporting.", "TIME_PERIOD", "OBS_VALUE")]
names(gender)[names(gender) == "Geopolitical.entity..reporting."] <- "country"
names(gender)[names(gender) == "TIME_PERIOD"] <- "year"
names(gender)[names(gender) == "OBS_VALUE"] <- "paygap_MF"
gender <- gender[-(125:134), ]
gender <- gender[-(82:100), ]
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```
#head(dataset)
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

For the sake of this example, I will continue with the assignment...

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
#inladen tidyverse
library(tidyverse)

#income_ineq aan het selecteren
income_ineq <- read.csv("data/general_income_ineq_(gini).csv") %>%
  select(geo, Geopolitical.entity..reporting., TIME_PERIOD, OBS_VALUE) %>%
  rename(
    country = Geopolitical.entity..reporting.,
    year = TIME_PERIOD,
    gini = OBS_VALUE) %>%
  filter(
    geo != "EA20",
    geo != "EU27_2020",
    geo != "Kosovo*")
```

```
education <- read.csv("data/tertiary_edu_govt_exp.csv")
education <- education[, c("geo", "Geopolitical.entity..reporting.", "TIME_PERIOD", "OBS_VALUE")]
names(education)[names(education) == "Geopolitical.entity..reporting."] <- "country"
names(education)[names(education) == "TIME_PERIOD"] <- "year"
names(education)[names(education) == "OBS_VALUE"] <- "edu_exp"
education <- education[-(109:119), ]
```

```
tot_gov <- read.csv("data/total_govt_exp.csv")
tot_gov <- tot_gov[, c("geo", "Geopolitical.entity..reporting.", "TIME_PERIOD", "OBS_VALUE")]
names(tot_gov)[names(tot_gov) == "Geopolitical.entity..reporting."] <- "country"
names(tot_gov)[names(tot_gov) == "TIME_PERIOD"] <- "year"
names(tot_gov)[names(tot_gov) == "OBS_VALUE"] <- "tot_exp"
tot_gov <- tot_gov[-(141:150), ]
tot_gov <- tot_gov[-(81:110), ]
```

```
library(tidyverse)
education$country <- NULL
rel_edu <- education %>% inner_join(tot_gov, by = c("geo", "year"))
rel_edu <- rel_edu %>% mutate(education_share = edu_exp / tot_exp * 100)
rel_edu$education_share <- round(rel_edu$education_share, 4)
avg_rel_edu <- rel_edu %>% group_by(geo, country) %>% summarise(avg_edu = mean(education_share), n_obs =
```

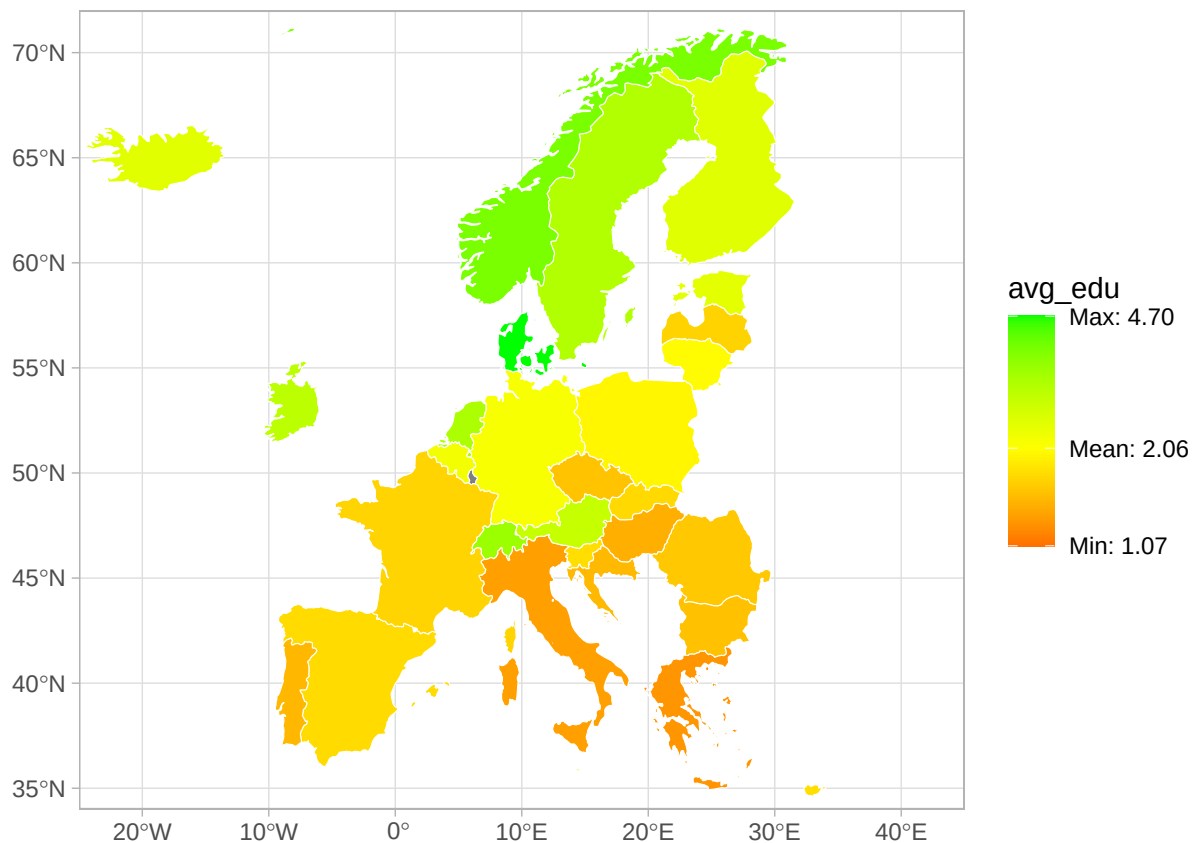
```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by geo and country.
## i Output is grouped by geo.
## i Use 'summarise(.groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(geo, country))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
#install.packages("sf")
#install.packages("rnatuarearth")
#install.packages("rnatuarearthdata")
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(rnatuarearth)

world <- ne_countries(scale = "medium", returnclass = "sf")
map_eu <- world %>%
  inner_join(avg_rel_edu, by = c("name" = "country"))
ggplot(map_eu) +
  geom_sf(aes(fill = avg_edu), color = "white", linewidth = 0.2) +
  scale_fill_gradient2(low = "red", mid = "yellow", high = "green",midpoint = 2.615328, limits = c(1.07, 4.70)) +
  coord_sf(xlim = c(-25,45), ylim = c(34,72), expand = FALSE) +
  theme_light() +
  labs(
  )
```



Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

```
#create dataset w gini avg (per country)
avg_gini_data <- income_ineq %>% group_by(geo, country) %>%
summarise (avg_gini = mean(gini), n_obs = n()) %>%
arrange(desc(avg_gini))
```

```
## 'summarise()' has regrouped the output.
## i Summaries were computed grouped by geo and country.
## i Output is grouped by geo.
## i Use 'summarise(groups = "drop_last")' to silence this message.
## i Use 'summarise(.by = c(geo, country))' for per-operation grouping
## ('?dplyr::dplyr_by') instead.
```

```
#join avg_gini and avg_edu in new dataset
library(tidyverse)
gini_edu_data <- full_join(avg_gini_data, avg_rel_edu, by = c("geo", "country"))
gini_edu_data$n_obs.x <- NULL
gini_edu_data$n_obs.y <- NULL
```

```
gini_edu_data$region <- replace(gini_edu_data$geo, gini_edu_data$geo %in% c("DK", "EE", "FI", "IE", "IS",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("AT", "BE", "CH", "DE",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("AL", "CY", "EL", "ES",
gini_edu_data$region <- replace(gini_edu_data$region, gini_edu_data$region %in% c("BG", "CZ", "HU", "PL",
```

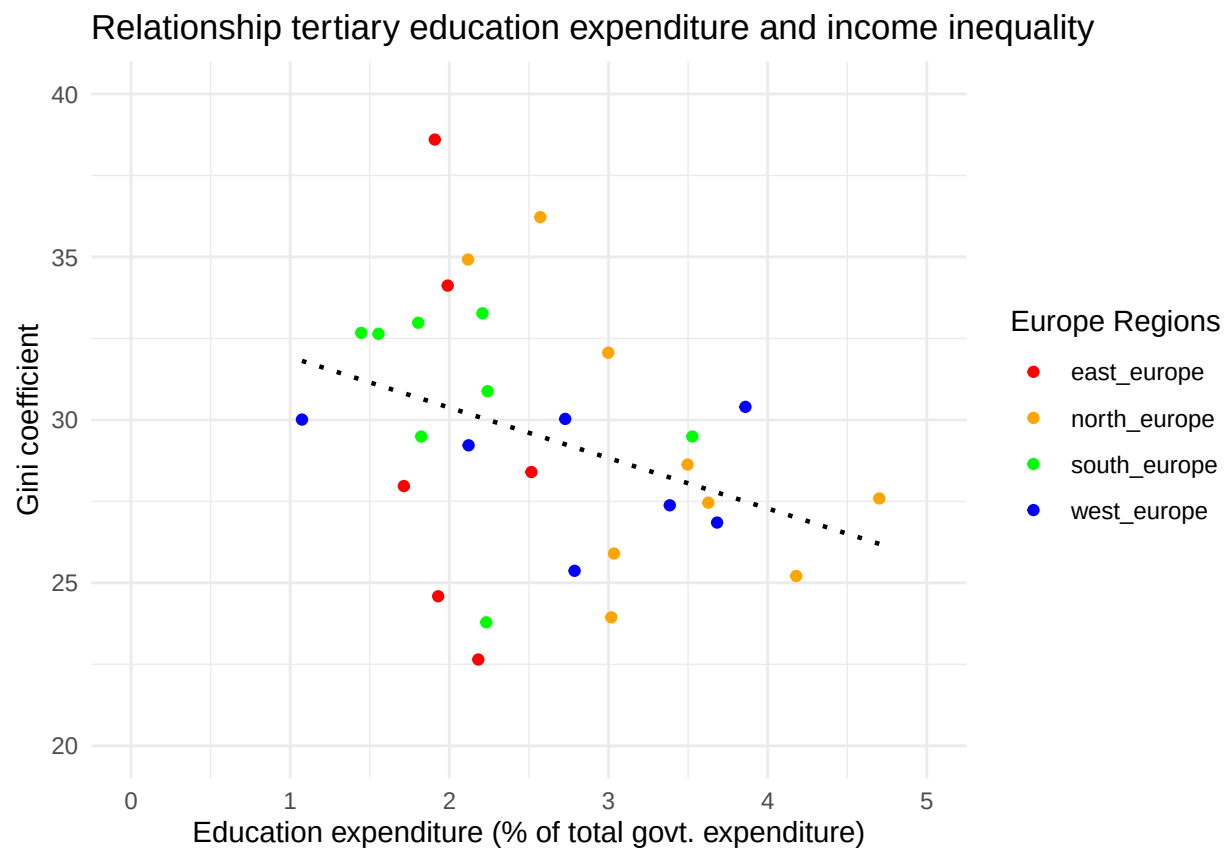
```
#creating scatterplot from gini_edu_data
ggplot(gini_edu_data,
      aes(x = avg_edu,
          y = avg_gini,
          colour = region)) +
  geom_point() +
  scale_colour_manual(
    values = c(
      "west_europe" = "blue",
      "north_europe" = "orange",
      "south_europe" = "green",
      "east_europe" = "red")) +
  scale_x_continuous(
    limits = c(0, 5)) +
  scale_y_continuous(
    limits = c(20, 40)) +
  labs(
    title = "Relationship tertiary education expenditure and income inequality",
    x = "Education expenditure (% of total govt. expenditure)",
    y = "Gini coefficient",
    colour = "Europe Regions") +
  geom_smooth( #regression line
    method = "lm",
    se = FALSE,
```

```
colour = "black",
linetype = "dotted",
linewidth = 0.8) + theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 7 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 7 rows containing missing values or values outside the scale range
## ('geom_point()').
```



#I might wanna add: deviation from regression line (se = TRUE?) / correlation strength

3.2 Generate necessary variables

Variable 1

Variable 2

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.

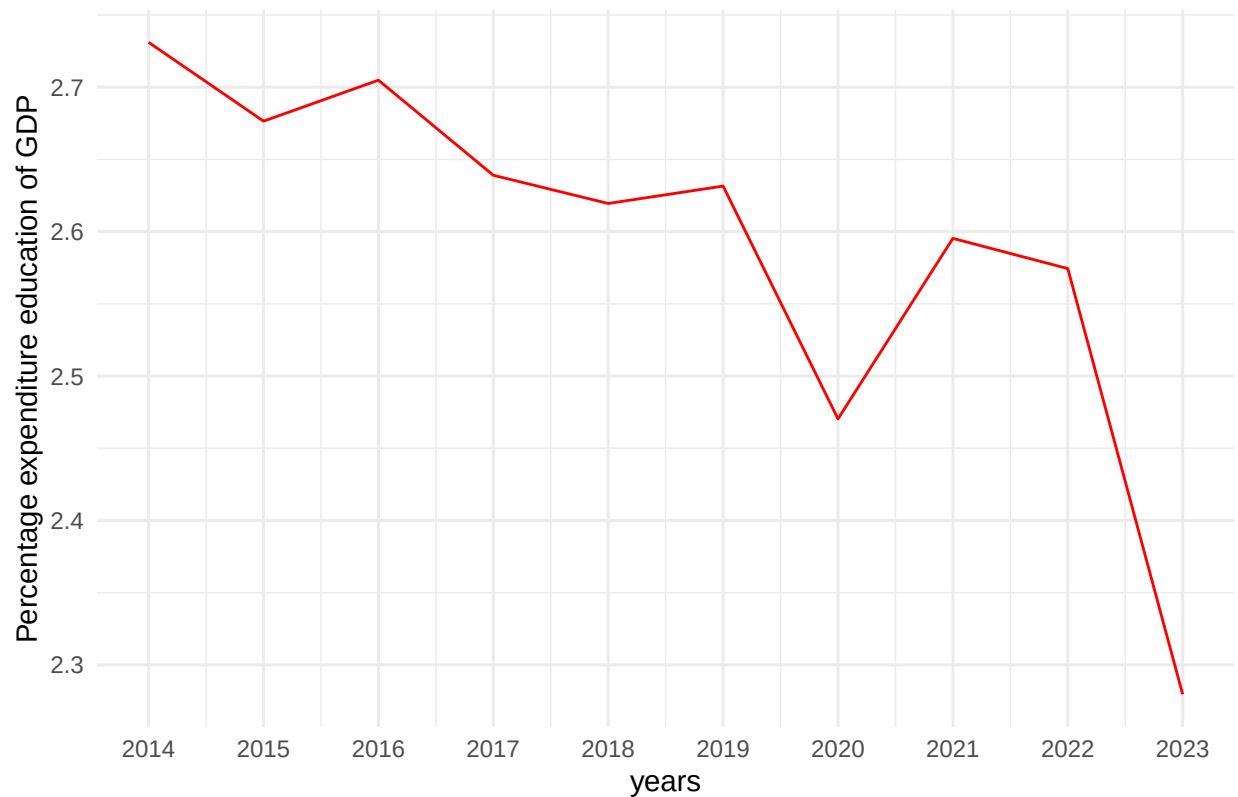
```
library(tidyverse)

year_avg_exp_eu <- rel_edu %>% group_by(year) %>%
  summarise(eu_ave_exp = mean(education_share),
            aantallLanden = n())

#maken van een grafiek met data (government expenditure over tijd)
library(tidyverse)
library(ggplot2)

year_avg_exp_eu %>%
  ggplot(aes(x=year, y=eu_ave_exp)) +
  geom_line(
    colour = "red") +
  scale_x_continuous(
    breaks = seq(2014,2023, by = 1)) +
  labs(
    title = "Average EU expenditure on education over the years (2014-2023)",
    x = "years",
    y = "Percentage expenditure education of GDP"
  ) +
  theme_minimal()
```

Average EU expenditure on education over the years (2014–2023)



```
#Graph average gini over the years EU

library(tidyverse)
gini_avg_pyear_eu <- income_ineq[income_ineq$country != "Kosovo*"],] %>% group_by(year) %>%
  summarise(avg_gini = mean(gini),
            aantallanden = n())

#Making of graph

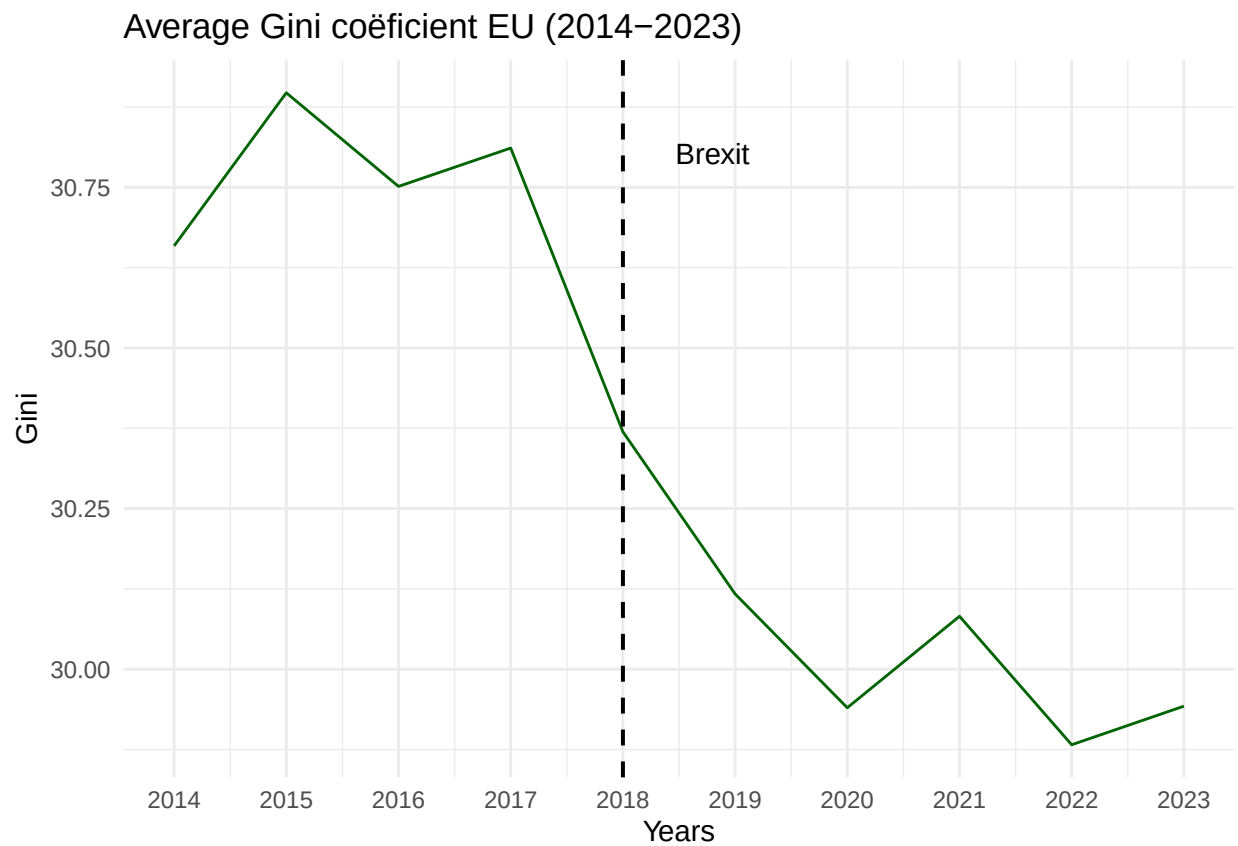
library(ggplot2)

gini_avg_pyear_eu %>%
  ggplot(aes(x = year, y = avg_gini)) +
  geom_line(
    colour = "darkgreen") +
  scale_x_continuous(
    breaks = seq(2014,2023, by = 1)) +
  labs(
    title = "Average Gini coefficient EU (2014-2023)",
    x = "Years",
    y = "Gini"
  ) +
  theme_minimal() +
  geom_vline(
    xintercept = 2018,
    linetype = "dashed",
```

```

    color = "black",
    linewidth = 0.7) +
  annotate(
    "text",
    x = 2018.8,
    y = 30.77,
    label = "Brexit",
    angle = 0,
    vjust = -0.5
  )
)

```



```

#graph komt wel sterk in aanmerking voor de shock
# Kosovo weggehaald voor de shock, er was maar één meting van 2018
# Shock komt door BREXIT

```